NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR
Department of Computer Science & Engineering | Course: CST037 Data Analytics

Data Analytics Project

Online Shopper Purchasing Intention Analysis

A Study on Sampling Dynamics, Statistical Hypothesis Testing, and Machine Learning Prediction

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1. DATASET OVERVIEW & PRE-PROCESSING

UCI Online Shoppers Purchasing Intention Dataset

12,330	18	15.47%	0
Total Sessions (N)	Original Attributes	Purchase Class (Revenue=1)	Missing Values

Attribute Classification

- **Numerical (10):** Page counts & durations (Administrative, Informational, Product Related), Bounce Rates, Exit Rates, Page Values, Special Day.
- **Categorical (8):** Month, Operating Systems, Browser, Region, Traffic Type, Visitor Type, Weekend, Revenue (Target).

Data Pre-processing Pipeline

- **Binary Encoding:** Target variable Revenue converted to binary Purchase indicator (1: Purchase, 0: No Purchase).
- **Data Splitting:** 80/20 Stratified Train-Test split (9,864 Training / 2,466 Test instances).

Simple Random Sampling (SRS) vs. Stratified Sampling

Fig 1: Purchase Probability Convergence (p = 0.1547)

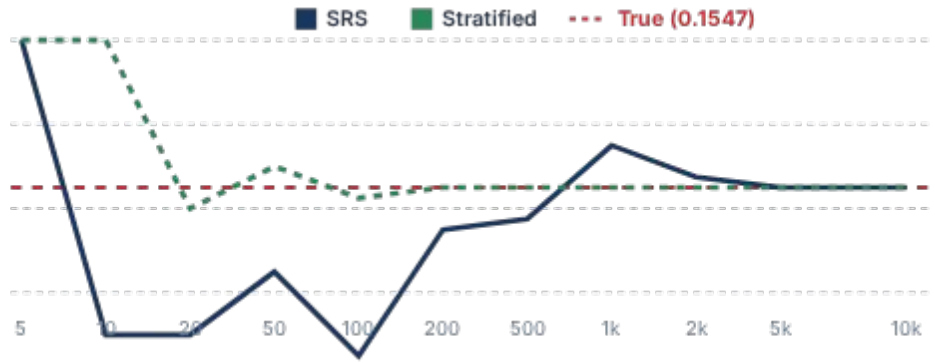
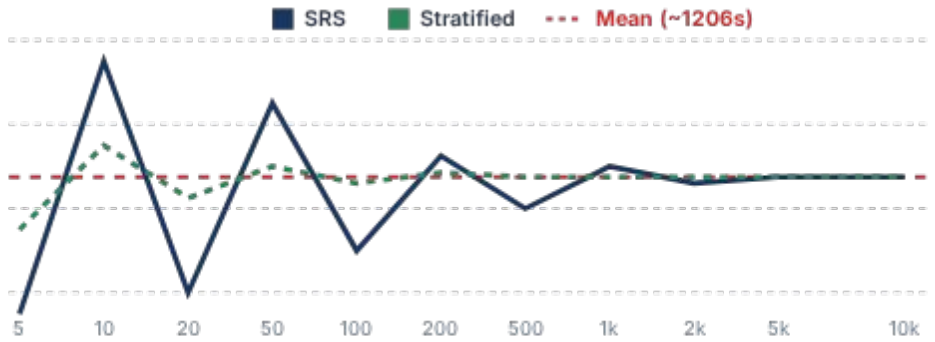


Fig 2: Mean Product Browsing Duration (μ ≈ 1,206 sec)



1. Small-Sample Volatility

At n ≤ 100, SRS mean duration fluctuates widely (±35%) due to extreme session length outliers in non-purchasers.

2. Stratified Variance Reduction

Stratification effectively minimizes sampling error across both class proportion (p=0.1547) and continuous browsing duration.

3. Large-n Convergence

Both schemes converge strictly to true population parameters (μ ≈ 1206s, p = 15.47%) once sample size reaches n ≥ 5,000.

3. STATISTICAL INFERENCE

Hypothesis Testing & Attribute Correlations

Welch's Two-Sample T-Test

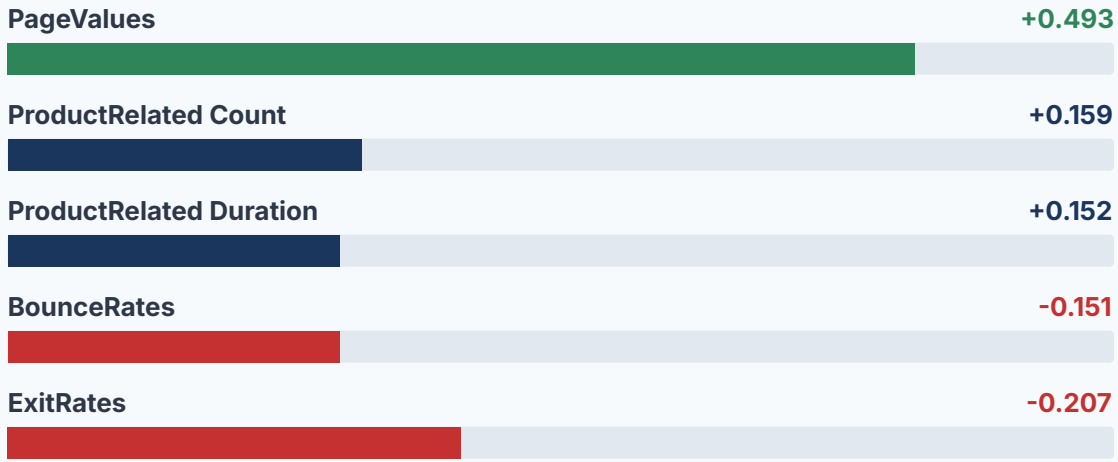
Testing: Product-Related Duration difference between Purchasers and Non-Purchasers.

$$H_0: \mu_{purchase} = \mu_{no_purchase}$$
$$H_1: \mu_{purchase} \neq \mu_{no_purchase} \quad (\alpha = 0.05)$$

Sample (n)	t-statistic	p-value	Result
50	1.9959	0.0963	**Fail Reject**
500	3.0101	0.0035	**Reject H0**
10,000	13.3565	5.88×10-39	**Reject H0**

Takeaway: Purchasers spend significantly more time on product pages. Larger sample size increases power to detect significant differences.

Attribute Correlation with Purchase Decision



* PageValues exhibits the strongest positive intent correlation. High Bounce & Exit rates strongly indicate non-purchasing behavior.

4. MACHINE LEARNING EXPERIMENT

Model Evaluation & Empirical Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest Classifier	**89.78%**	72.41%	54.97%	**0.6250**	**0.9179**
K-Nearest Neighbors (k=3)	86.50%	60.00%	38.48%	0.4689	0.7469

K-Fold Cross Validation

A **4-fold cross-validation** strategy was implemented to select the optimal neighbor parameter (k = 3) for KNN, preventing overfitting on training splits.

Why Random Forest Excelled

Random Forest effectively handles non-linear relationships, mixed tabular data types, and sparse dummy variables better than distance-based metrics (KNN).

Minority Class Challenge

Despite a high ROC-AUC (0.9179), Random Forest recall (54.97%) reflects the difficulty of capturing all transactions in an imbalanced dataset (15.47% positive).

Reference Paper: Qi Jiang (Wenzhou Kean University, China)

Purchase Intention Analysis Based on ML & K-Means Clustering

1. Classification Model Comparison

Evaluated three predictive classifiers on online session data:

Random Forest Best Performer
Achieved best overall Accuracy, F1-Score, and Recall.

XGBoost
Strong gradient boosted decision tree baseline.

Artificial Neural Network (ANN)
Higher computational complexity without outperforming RF.

Takeaway: Confirms Random Forest as the superior classifier for tabular e-commerce session data.

2. K-Means Customer Behavioral Segmentation (K = 4)

Cluster 0 — Frequent Browsers
Low Interest / High Frequency
Users browse frequently but show no clear purchase intention.

Cluster 1 — Deep Engagers
Strong Purchase Intention
Users demonstrate strong interest and deeper product engagement.

Cluster 2 — Quick Exits
High Exit Rate
Low product-related activity with rapid session departure.

Cluster 3 — High Potential
High-Potential Buyers
Strong engagement and low exit rates, signaling high conversion potential.